

• World Sheet coordinates  $\xi^\alpha$ ,  $\alpha = 0, 1$

Spacetime indices:  $A, B, \dots$

$$\eta_{\alpha\beta} = (-1, 1), \quad \gamma^B = (i\sigma^2, \sigma^1) \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\{\gamma^\alpha, \gamma^\beta\} = 2\eta^{\alpha\beta}$$

$$\sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\alpha, \beta$ : World sheet  $\Sigma$   
 $A, B$ :  $\text{Spin}(\Sigma, n)$   
 $M, N$ :  $\mathbb{M}^D \hookrightarrow \Sigma_2$

$$\gamma_5 = +\gamma^0\gamma^1 = i\sigma^2\sigma^1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma^3$$

$$\gamma_5 = \sigma^3$$

• Charge Conj:  $C\gamma^B C^{-1} = -\tilde{\gamma}^B$

$$C_{AB} = -\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\gamma^0$$

• Dirac Conj:  $\bar{\chi}^D = -\chi^\dagger \gamma^0$

$$\text{Charge conj: } \chi^C = C\bar{\chi}, \quad \bar{\chi}^C = C^{-1}\chi = \tilde{\chi}C$$

\* Chiral mult.  $(A, \chi, N)$

$$A = \frac{1}{2} \int d^2\xi \left( -\partial_\alpha A \partial^\alpha A - i \bar{\chi} \not{\partial} \chi + N^2 \right)$$

$\rightarrow$  Invariant under Majorana  $(1, 1)$  SUSY

$$\delta A = i \bar{\epsilon} \chi, \quad \delta \chi = \not{\partial} A \epsilon + N \epsilon$$

$$\delta N = i \bar{\epsilon} \not{\partial} \chi$$

$A, N$ : real scalar fields

$\chi$ : Majorana spinor

$\rightarrow$  Couple this to the supergravity multiplet  $(\psi_{\alpha\beta}, \varphi_{AA})$

which has its own action: free

Rarita-Schwinger Action

$$S = - \int d^2x \bar{\varphi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \varphi_\rho$$